HW 7

Due on 10 April 2026 at 8PM on Gradescope.

This assignment is worth 20 points, and you should be spending roughly 10 minutes per point on average. If you find yourself spending more than $10n$ minutes on a n-point problem, reach out to me via drop-in hours, Discord, or email.

Collapse all exercises

Goals

- Practice thinking about curvature
- Work on your blog post
- Extra credit opportunity

▼ EXERCISE 1 [2]

1. Fill out the following and tag it as Exercise 1:

```
\section*{Exercise 1}
Collaborators: Alex, Ross, Margaret, Joan

\noindent
Tools and resources used: On Exercise 1, I used ChatGPT to help
\pagebreak
```

You don't need to include class notes or textbooks under "tools and resources."

2. On *each* individual exercise you attempt (both required and choice) start with a header based on this template. (Make sure that the denominator of the "self-assessed score" matches the number of points attempted.)

```
\section*{Choice Exercise 5}
\begin{itemize}
  \item Points attempted: 7
  \item Self-assessed score: 6/7.
  \item Time spent: 56 minutes.
  \item Difficulty: 8/10.
  \item Questions/Feedback: None.
\end{itemize}
```

3. Choose one or more problems that you would most like feedback on, and indicate this under the "questions/feedback" bullet with a brief note of what

you would most like me to focus on. (This could be "mathematical correctness," "grammar," "notation," "formatting," or anything else.)

▼ EXERCISE 2 [1] (DISCORD)
Post on Discord with a hint or question about your blog post, markdown, Exercise 3, or any previous homework exercise.

▼ EXERCISE 3 [7] (CURVATURE)

Let M be the surface given by the parametric equation

$$\mathbf{r}(u,v) = \begin{pmatrix} u \\ v \\ uv \end{pmatrix}, \quad (u,v) \in (-1,1) \times (-1,1),$$

and let $\mathbf{N}\colon M \to S^2$ be the Gauss map, where

$$S^2 = \{(x,y,z) \mid x^2 + y^2 + z^2 = 1\}$$

denotes the unit sphere.

We will be calculating the various notions of curvature at $p = \mathbf{r}(0,0)$ $= (0,0,0)$.

1. We begin by computing the Gauss map at each point and its differential, which we denote $dN_p\colon T_pM \to T_{N(p)}(S^2)$, and which is defined at each point $p \in M$.

- **a.** Begin by computing $N_p\colon M \to S_2$ at each point $p = \mathbf{r}(u,v) \in M$ in terms of u and v.
- **b.** Next, explain why the tangent space T_pM at each point p is identical to the tangent space $T_{N(p)}(S^2)$ by comparing the two tangent planes geometrically.
- **c.** Compute the differential (Jacobian matrix) $dN_p\colon T_pM \to T_{N(p)}(S^2)$ with respect to u and v, and evaluate it at $(u,v) = (0,0)$.

2. Next, you will guess at what this map looks like:

- **a.** Use a computer to plot M around the point $p = \mathbf{r}(0,0) = (0,0,0)$.
- **b.** Compute $\mathbf{r}_u(0,0)$ and $\mathbf{r}_v(0,0)$.
- **c.** Describe in words what $dN_p(\mathbf{r}_u(0,0))$ and $dN_p(\mathbf{r}_v(0,0))$ represent.
- **d.** By looking at the plot, make a guess of what you expect to get after you compute $dN_p(\mathbf{r}_u(0,0))$ and $dN_p(\mathbf{r}_v(0,0))$.

e. Now compute these values. If your guess was wrong, figure out why, and explain where your intuition went wrong.

3. Compute the eigenvalues of dN_p at $p = \mathbf{r}(0,0) = (0,0,0)$ and their corresponding eigenvectors. Prove that the eigenvectors are orthogonal to each other. *(The (unit) eigenvectors are called the principal directions and the (negative) eigenvalues are called the principal curvatures at p.)*

4. The *mean curvature* is the mean of the principal curvatures. Compute it.

5. The *Gaussian curvature* is the product of the principal curvatures which is the determinant of dN_p. Compute the Gaussian curvature.

6. We say that a point p on a surface is

a. planar if dN_p is the zero matrix,

b. hyperbolic if $\det(dN_p) < 0$,

c. parabolic if $\det(dN_p) = 0$ but dN_p is nonzero, and

d. elliptic if $\det(dN_p) > 0$.

How can we characterize the point p on M according to this classification?

7. Draw or plot a smooth surface S where some points are elliptic, some are parabolic, and some are hyperbolic. Include the image here.

▼ EXERCISE 4 [10] (BLOG POST)

Spend at least 100 minutes working on your blog post in markdown. Save the markdown file as `blog_post.md` and include it in your LaTeX document with `\usepackage[settings]{markdown}`. (You can read more about this here.)

Extra Credit

You can do up to five points of extra credit, which you can claim on any previous homework. After completing up to five points of the problems below, request a regrade on the "final choice exercise" of the homework assignment for which you want to apply the extra credit points.

▼ FINAL CHOICE EXERCISE [≤5]

Do any subset of the following parts, state which you did and how many points you are claiming for this problem (maximum of 7 points).

1. [≤3] Email Prof Kagey (pkagey@cpp.edu) or post in the `#feedback` channel on Discord with a typo that you have spotted in this assignment or with a suggestion about how to make something clearer. (One point for every six minutes that the typo got you stuck, with a maximum of three points.)

2. [≤2] Ask a question on Discord. (One point per question, up to 2.)

3. [≤4] Answer a classmate's question on Discord or give a hint about a problem on this week's assignment.* (Two points per answer/hint, up to 4.)

4. [≤4] Can you generalize any of the above problems in an interesting way? If so say what the generalization is, attempt to solve it, say something about it, and say how many points you feel like your effort was worth, up to 4.
5. [≤4] Choose one or more problems from our textbook or a previous homework assignment which interest you and write the problem statement along with an answer. Say how many points you feel like your effort was worth, up to 4.
6. [2] Stop by student drop-in hours.
7. [2 or 5] Work on at least one of the above exercises with a classmate. State which classmate you worked with and which exercises you collaborated on. If you've collaborated with someone for the first time, you get 5 points, if you've collaborated with someone before, you get 2 points for working with them again.
8. [2] Turn in the assignment on time (without requesting an extension).

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[1] The syntax for a hint on Discord is double bars `||` on each side of the spoiler like this: `Remember that ||1 + x < x + 2||.` ↩

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